නව නිර්දේශය/ புதிய பாடத்திட்டம்/New Syllabus

NEW

කොළඹ දිස්ත්‍රික් පූර්ව වෛද්‍ය ශිෂ්‍ය සංගමය 2021
கொழும்பு மாவட்ட முன் மருத்துவச்சங்கம் 2021
Pre Medicine Association, Colombo District 2021

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය (ආදර්ශ), 2022
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை (மாதிரி), 2022
General Certificate of Education (Adv. Level) Examination (Model), 2022

භෞතික විද්‍යාව II
பௌதிகவியல் II
Physics II

01 S II

පැය තුනයි
மூன்று மணித்தியாலம்
Three Hours

අමතර කියවීම් කාලය - මිනිත්තු 10 යි
மேலதிக வாசிப்பு நேரம் - 10 நிமிடங்கள்
Additional Reading Time - 10 minutes

Use add reading time to go through the question paper. Select the questions you will answer and decide which of them you will prioritise

Index no:- ....

Important:

- *This question paper consists of* **16** *pages.*
- *This question paper comprises of* ***two*** *parts, Part* **A** *and Part* **B**. *The time allotted for* ***both parts*** *is* ***three hours***.
- *Use of calculators is* ***not*** *allowed.*

PART A — Structured Essay:
(pages 2 - 8)

Answer ***all*** *the questions on this paper itself. Write your answers in the space provided for each question. Note that the space provided is sufficient for your answers and that extensive answers are not expected.*

PART B — Essay:
(pages 9 - 16)

This part contains ***six*** *questions, of which,* ***four*** *are to be answered. Use the papers supplied for this purpose.*

- *At the end of the time allotted for this paper, tie the* ***two parts together*** *so that* ***Part A*** *is* ***on top*** *of* ***Part B*** *before handing them over to the Supervisor.*
- *You are permitted to remove* ***only Part B*** *of the question paper from the Examination Hall.*

For Examiners' Use Only

For the second paper		
Part	**Question Nos.**	**Marks Awarded**
A	1	
	2	
	3	
	4	
B	5	
	6	
	7	
	8	
	9 (A)	
	9 (B)	
	10 (A)	
	10 (B)	
Total	In numbers	
	In words	

Code Numbers

Marking Examiner 1	
Marking Examiner 2	
Marks checked by	
Supervised by	

[දෙවැනි පිටුව බලන්න]

01) The diagram given below shows an incomplete experimental setup to determine the coefficient of viscosity of a liquid by capillary flow method.

a) i) Complete the above diagram and name them properly.
(Mark the direction of water flow with arrows.)

ii) Write Poiseuille's formula according to the measured quantities and rearrange the equation to get a straight-line graph to find the coefficient of viscosity.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

iii) Identify and name thephysical quantities of all the symbols used in the rearranged equation.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

iv) Which physical quantity should be measured more accurately? State a more accurate way of measuring it rather than using a measuring instrument and the formula with common symbols to find it.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

b) i) State the experimental steps to obtain values for variables, in order.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

ii) Write down the procedures to be done on the capillary tube and reasons for doing so, before and during setting up of the capillary flow apparatus.

	Experimental Precautions	Reason
Before setting up		
When setting up		

c) i) Give the two factors which affect viscosity in this experiment and state how it is nullified.

………
………
………
………

ii) Give 2 reasons why a string is attached to the end of the capillary tube?

………
………

d) i) A non-straight line graph pattern is observed after a particular height when increasing the height of the water level in constant pressure head from the capillary tube. What could be the reason for the change in the shape of the graph?

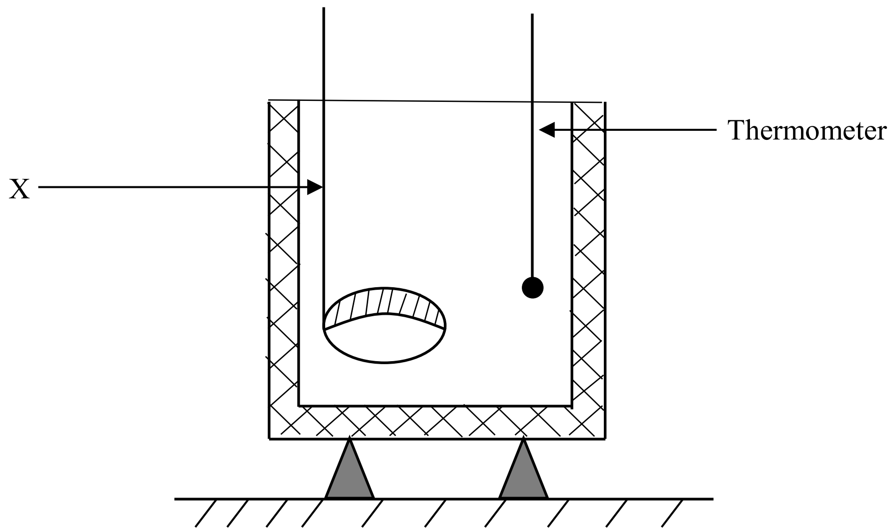
………
………
………

ii) It is observed that volume of liquid collected at a particular time is not sufficient to take a measurement using the instrument even though using the whole length of the rubber tube. State a method to rectify this without changing the physical quantity of the instrument and maintaining the same time period as earlier.

………
………

iii) Draw a sketch of the graph obtained in this experiment including the scenario observed in (d) (i) and after rectifying scenario (d) (ii).

02) The diagram below shows an experimental setup that is used to find the specific latent heat of fusion of ice.



a) In order to conduct the above experiment, water which is at a temperature higher than that of room temperature, should be added to calorimeter. Indicate by drawing on the above diagram, the level up to which water should be added into the calorimeter.

b) Name X. What is the reason for using X for the above experiment?

………
………
………

c) The readings which should be taken in this experiment can be denoted as a,b,c,d,e as shown below. Name the readings in order. Mention whether these readings are taken before or after adding ice to water.

a - ……………………………………………………………………………………………

b - ……………………………………………………………………………………………

c - ……………………………………………………………………………………………

d - ……………………………………………………………………………………………

e - ……………………………………………………………………………………………

d) Write down an expression for the specific latent heat of fusion of ice, if the specific heat capacity of ice is C_w and the specific heat capacity of the calorimeter (with stirrer) is C_1. (In terms of a,b,c,d,e)

………
………
………
………
………
………
………

e) In this experiment

1. What is the purpose of using of ice that is melting?

………
………

2. Why is powdered ice not used?

……
……
……
……………………………………………………………………………………………………..………….

3. Why should the content be stirred quickly?

……
……
……

f) How does the value of L obtained from the experiment, deviate from the actual value of L, if a small amount of water is spilled out from the calorimeter

……
……
…….

g) How does the atmospheric pressure affect this experiment ?

……
…….

h) If dew is formed on the outer surface of the container, with the value of L found experimentally be greater than or less than the actual value of L? What is the reason for your answer?

……
……
……
……

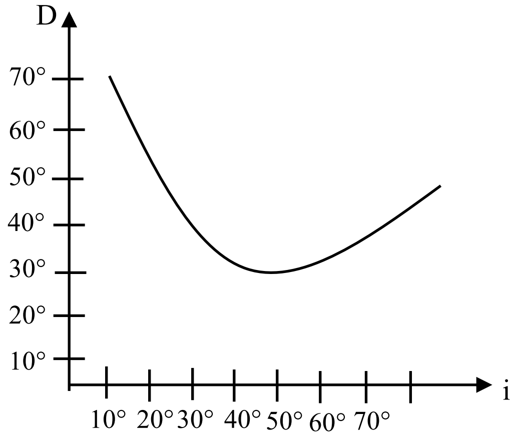
i. A 2kg mass of water at $70°$C is kept inside a refrigerator until it becomes ice at $-11°$C. During this process 1302.2kJ of energy is removed by the refrigerator. If the specific heat capacity of water is $4200\ \mathrm{Jkg^{-1}°C^{-1}}$ and the specific heat capacity of ice is $2100\ \mathrm{Jkg^{-1}°C^{-1}}$, Find the specific latent heat of ice.
(Neglect the heat loss to the surrounding)

……
……
……
…….
……
……
……
…….
……
……
………..

03) A)

i. Draw a ray diagram showing the instance of minimum deviation of a ray of light travelling through a prism and mark all angles.

ii. Given below is a graph plotted using the readings obtained by a student from an experiment conducted to find the angle of minimum deviation of a prism.



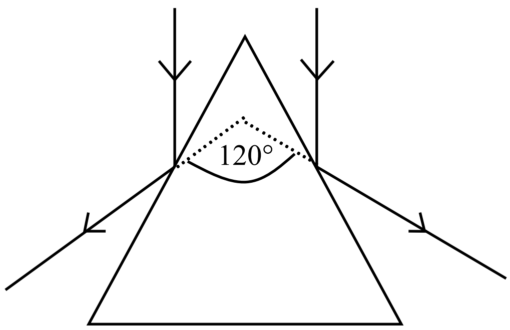
Find the angle of minimum deviation of the prism using the graph.

....

iii. The ray diagram obtained by the student when finding the prism angle of the prism using the spectrometer, is shown below.

Hence, calculate the prism angle of the prism.

....



iv. Write down the equation that is used to find the retractive index (n) of the material of the prism, in terms of the angle of minimum deviation of the prism (d_{min}) and the prism angle (A).

....

....

v. Accordingly, calculate the retractive index of the material used to make prism.

………
………
………...

b) A student has designed an experiment to find the retractive index of water relative to air, using the critical angle method.
For this in order to first determine the critical angle of the glass relative to air the student has placed a glass prism on a white piece of paper as shown in diagram 1 below. Here, the pin P is place such that it is in contact with the face xy of the prism.

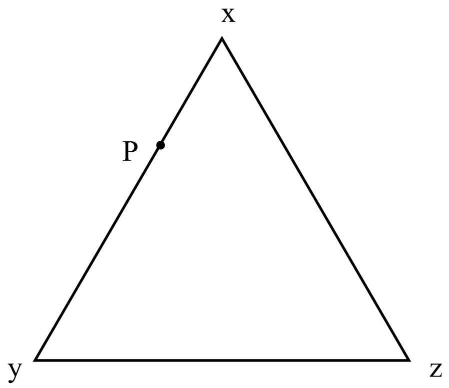


Diagram 1

i) What is the reason for placing pin P to be in contact with the face xy?

....

ii) How is the path of the emergent ray experimentaly determined using 2 more pins?

....

....

iii) Write down the relationship between the critical angle of the glass with respect to air (C_1) and the refractive angle of glass with respect to air (n_1)

....

....

c) The student, after determining C_1 using the proper experimental procedure, conducted the experiment once again as before by placing a thin layer of water on the xz surface. The position of P is not changed in this occasion as well.

i) What is the new location of the image of P relative to the previous location of the image of P.

....

....

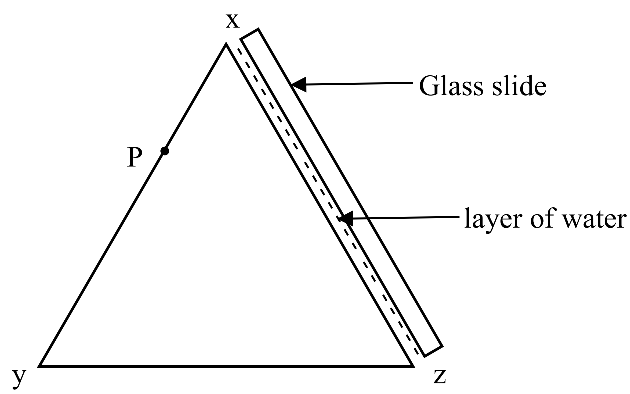


Diagram 2

ii) The student has determined that the critical angle for the interface of glass-water (C_2). Write down an expression for the retractive index of water with respect to air (n_2) , using C_1 and C_2.

....

....

....

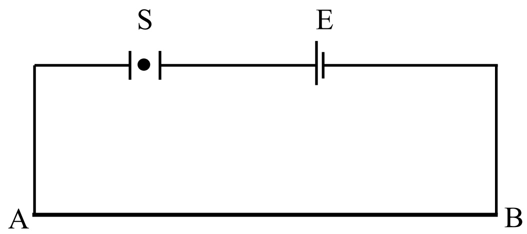
iii) If values obtained for C_1 and C_2 are $4°50'$ and $6°30'$ respectively, calculate the value of n_2.

....

....

....

04) A) The diagram below shows a potentiometer circuit.



Here, E represents the dricer cell, S represents the switch and AB represents a potentiometer wire of length 10m.

1) i) What is the type of cell used as the driver cell?

....

ii) What is the special advantage of using that particular cell as the driver cell?

....

2) What is the metal that is used to make the potentiometer wire and mention any special reason for selecting that metal.

....

....

....

3) i) State the type of key used as switch S.

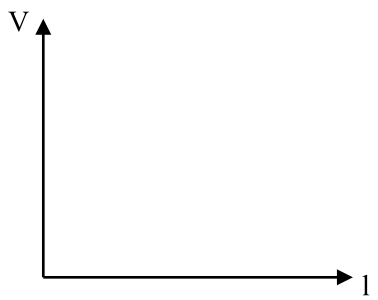
....

ii) Mention the reason for your selection.

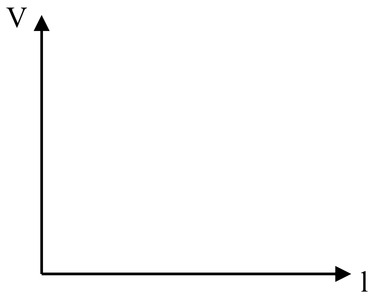
....

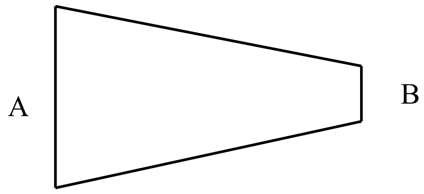
....

4) Plot a graph showing the variation of the potential drop across the potentiometer wire arranged here with the legth of the potentiometer wire.



5) If, by any means the potentiometer wire is made in such a way that the cross section area of it decreases uniformly as shown in the diagram, plot the variation of the potential with the distance.

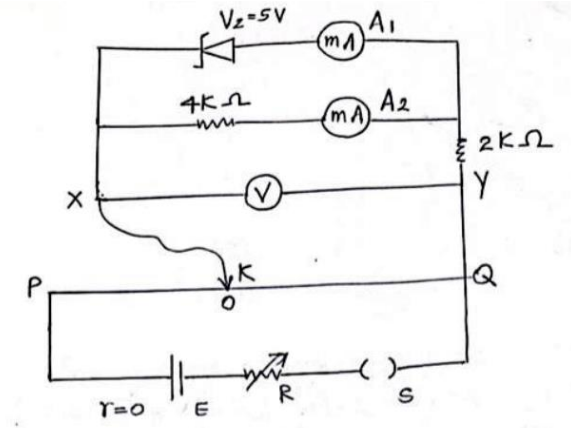




6) What is meant by 'Calibrating a potentionmeter' ?

……

B) The following circuit is contructed to find the nature of action of the Zener diode. The voltmeter and the ammeters used are ideal. The Zener voltage is 5V.



The touch key connected to end X is placed such that it is in contact with the potentiometer wire PQ is measured from end Q and $QO = l$

i) When the touch key is in contact with the potentiometer wire, is the Zener diode forward biased or reverse baised ?

……

ii) Calculated the readings of ammeters A_1 and A_2 when the voltmeter reading is 6V.

……

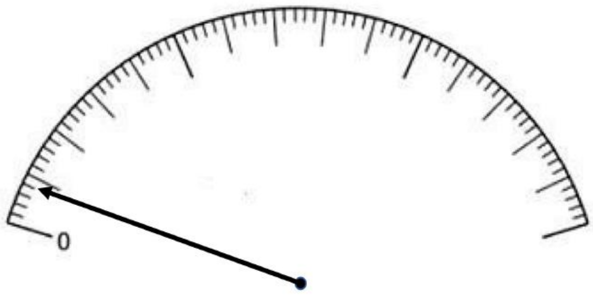
……..

iii) Calculate the readings of ammeters A_1 and A_2 when the voltmeter reading is 9V.

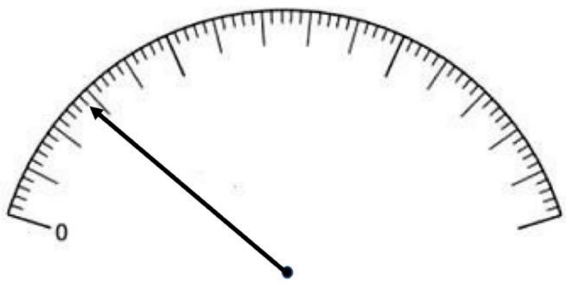
……

……..

iv) The figure (1) shows the position of the indicator when there is no current flow through the ammeter A_2 and figure (2) shows the indicator position when the current calculated in (ii) above flows through ammeter A_2.



(1)



(2)

A. Draw the position of the indicator on the figure (2) given above, if a current that is equivalent to the current that passes through the $2k\Omega$ resistor in instance (iii) given above, passes through the ammeter.

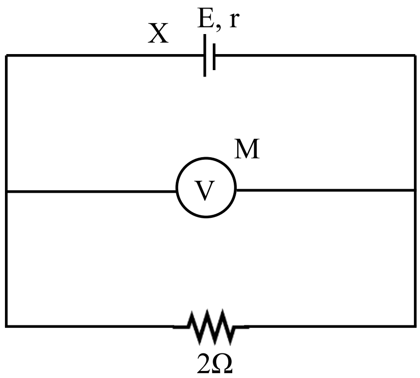
B. What is the least count of the ammeter?

……….

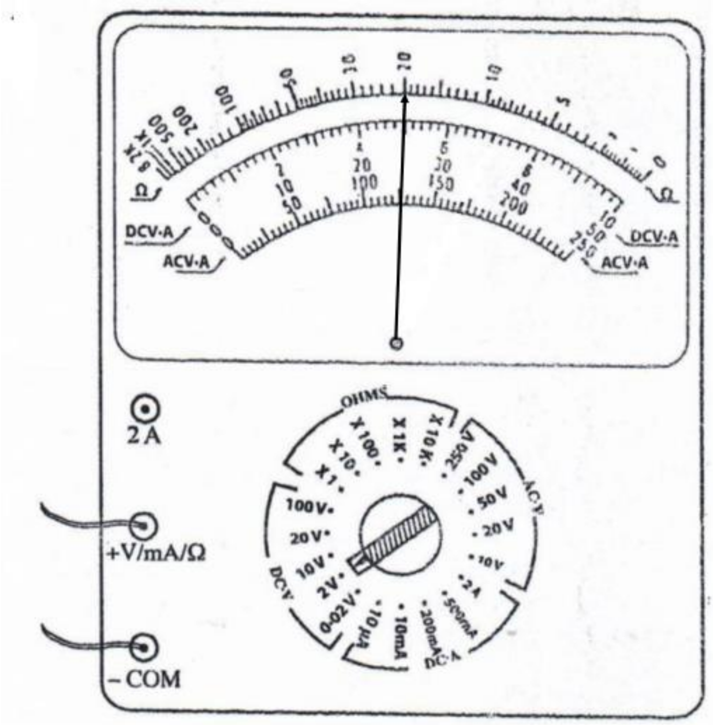
C. What is the zero error of the ammeter?

…………...

(C) The internal resistance (r) of a certain dry cell X was found to be 1Ω when an experiment was conducted using the potentiometer setup. Now, it is required to find the EMF E of the dry cell X. A simple circuit arrangement used for this purpose is shown below. M is an analoge multimeter that is used to obtain the terminal potential difference of cell X.



If the diagram below shows the reading shown by the multimeter M, find the E.M.F of dry cell X.



………
………
………
………
………
………

Essay

05)

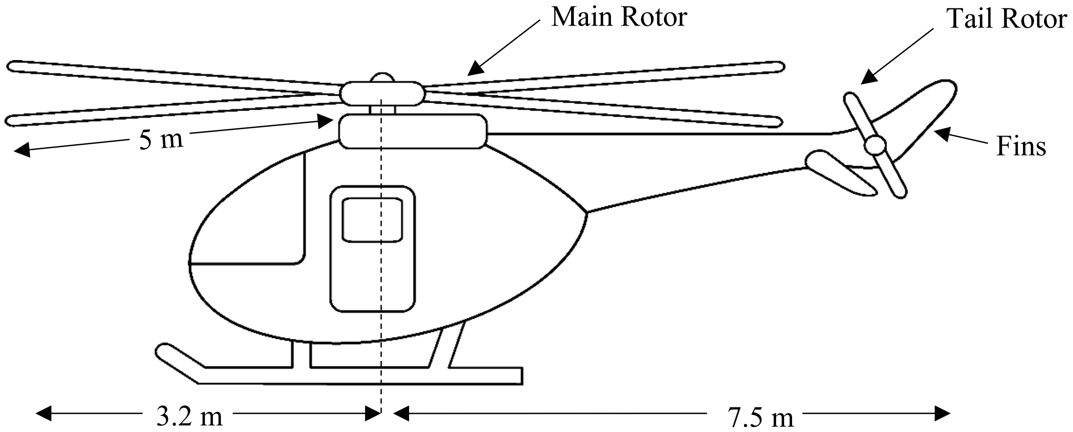


Figure 1

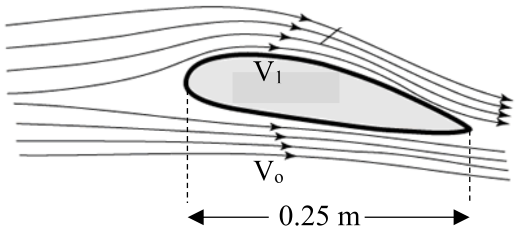


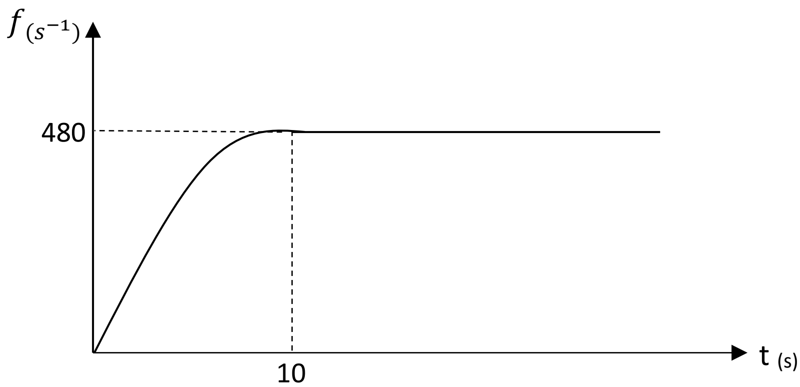
Figure 2

Figure 1 shows a helicopter and its accurate measurements. The mass of the helicopter is 960kg.

Figure 2 is a streamline diagram which shows the flow of air currents across a cross section of a main rotor blade of the helicopter.

a) 1.) Explain the Bernoulli's principle.
2.) Write down the Bernoulli's equation and identify its terms.
3.) Explain using the Bernoulli's principle, how the upward force required to lift the helicopter is produced by the main rotor blades of the helicopter.

b) Given below is a graph which shows the variation of the frequency of the main rotor blades of the helicopter with time. The angular velocity of the rotor blades is gradually increased until the helicopter is just lifted from the ground and after that the angular velocity is kept constant. The force generated by the tail rotor is 500N. The main rotor blades rotate in the clockwise direction.



1.) Calculate the velocity at a point which is situated 2.5m away from the axis of rotation of the main rotor blade.

2.) Consider the velocity calculated in 1 above, as the velocity of air currents travelling below the rotor blades, relative to the rotor blades. Now, using the Bernoulli's principle calculate the mean velocity of the air currents above the rotor blades, when the helicopter is just lifted from the ground. (Density of air- 1.2 kgm^{-3} and π^2=10)

3.) At the instance where the helicopter is just lifted from the ground, the body of the helicopter has a tendency to rotate in the anticlockwise direction. Explain the reason for this. What is the strategy used in the helicopter to prevent this anticlockwise direction?

4.) At the instance where the helicopter is just lifted from the ground, calculate the power of the main rotor blade against friction and air resistance.

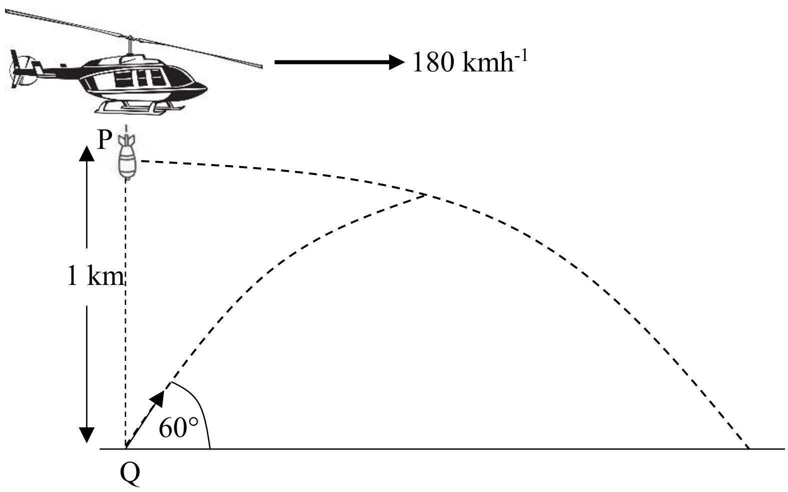
c) Consider a situation where the helicopter is flying in the upper sky at a constant velocity, with 30° inclined to the vertical. The air resistance force acting on an object can be given by the equation $F = 1/2k\rho v^2$. 'k' is a constant that depends on the object. It depends on the shape of the object and the surface area that is in contact with air.
($\rho = 1.0$ kgm^{-3})

1) Calculate the velocity of the helicopter in kmh^{-1} ($k = \sqrt{3}$)

2) Helicopter is flying with same speed calculated in 1 above. Now a heavy rope is hanging down with one end attached to the helicopter. Draw the equilibrium position if the rope. (Consider air resistance)

3) The rope is inclined 30° to the vertical when an object of mass 50kg is attached to its free end. Calculate the constant 'k' for this object, assuming the helicopter travels with the same constant velocity calculated in 1 above. (mass of the rope and the air resistance can be neglected compared to the object.)

d) While the helicopter is flying horizontally at a constant velocity of 180 km h^{-1}, a bomb 'P' is released from the helicopter C. (The bomb is released from rest relative to the helicopter). At the same time a bullet 'Q' is released from a point on the ground which is situated 1km vertically underneath the helicopter. The bullet Q is projected with an angle 60° to the horizontal.



1. Find the velocity the bullet 'Q' should have in order to collide with P
2. Find out the time taken for the collision to occur.
3. Calculate the horizontal distance that Q has travelled until the collision.

06) The audible range of frequencies is 20 Hz- 20 kHz. The sound waves with frequency higher than the 20 kHz are known as ultrasound waves and the sound waves with frequency less than 20Hz are known as infrasound waves.

a) i) State whether the propagation of sound waves in air is an adiabatic process or an isothermal process.

ii) Name 2 main energy forms by which sound waves propagate energy in air.

iii) What is meant by the intensity of a sound wave?

b) The intensity of a sound wave can be given by the equation $I = 2\rho VA^2\pi^2f^2$

ρ - (density of air) = 1.35 kgm^{-3}

V - (velocity of sound in air) = 300 ms^{-1}

A – the maximum displacement of atoms of molecules in equilibrium position

f - frequency of sound wave

i) Threshold of pain for the human ear is 120dB. Calculate the intensity relevant to this intensity level.

ii) A sound wave has an intensity level of 120dB and a frequency of 1000Hz. Calculate the maximum displacement of an air molecule. (π^2 =10)

iii) A sound wave has a frequency of 1000Hz and an intensity of 0.5 Wm^{-2}. Calculate the maximum displacement of an air molecule. ($\boldsymbol{\pi}^2$ = 10)

iv) If the pressure change in air during the propagation of a sound wave is P_0, it can be expressed as $P_0^2 = 2ZI$. Z is the acoustic impedance. Z = AV. If the maximum pressure change in b.) (ii) and b.) (iii) above are P_O^1 and P_O^2 respectively, Calculate P_{01} and P_{02}.

v) For a wave of frequency 1000Hz and intensity level 120dB,

1) Plot a graph to show the variation of displacement of air molecules with time.
2) Plot a graph to show the variation of pressure change in air with time.
3) If the air density changes during the propagation of the sound wave, plot a graph to show the variation of air density with time.

vi) For a wave of frequency 1000Hz and intensity 0.5 Wm^{-2}

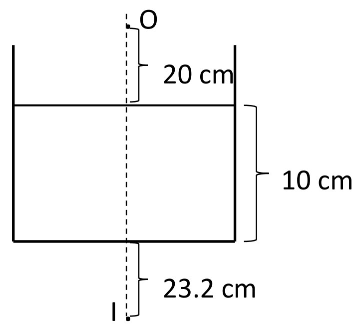
1) Plot a graph to show the variation of displacement of air molecules with time, in the same axis used for v) 1). Mark it as vi) 1.)
2) Plot a graph to show the variation of pressure change of air with time, in the same axis used for v) 2.) Mark it as vi) 2.)
3) Plot a graph to show the variation of density of air with time, in the same axis used for v) 3.) Mark it as vi) 3.)

c) i) The velocity of ultra sound waves in a gas can be given by $V = \sqrt{(P/\rho)}$. Here P is the pressure of the gas and ρ is the density of the gas. If the velocity of sound waves in a gas is u, show that $u = \sqrt{(\gamma V)}$. γ is the ratio of principle molar heat capacities of the gas.

ii) γ for air is 1.44. The speed of sound waves in air is 360ms^{-1}. Calculate the speed of ultrasound waves in air.

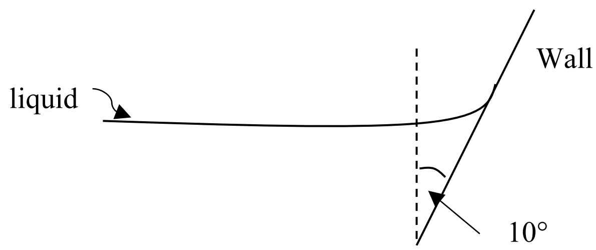
d) According to the following diagram, water is filled to a container up to 10cm. A plane mirror is kept at the bottom of the container. A tennis ball is kept 20cm above the water level in the container. When you look in to the container from above, the last reflection of the ball can be seen 23.2cm behind the plane mirror. (Consider the ball as a point object and the water surface is still)

i. How many reflections are created here?
ii. Draw a ray diagram to show the formation of the reflections.
iii. Find the refractive index of water

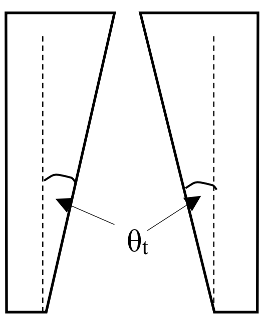


07) a)

i) What is the vertical force exerted by a fluid of T= 7.2 x 10^{-2} Nm^{-1} and contact angle 60°, on **a unit length** of a wall which is inclined by 10° to the vertical?



ii) The vertical cross section of a narrow capillary tube is shown above. Obtain an expression for the capillary rise (h) which occurs when this capillary tube is dipped in a liquid of surface tension coefficient T, density ρ and a contact angle θ_c with the glass.
Take gravitational acceleration as g and the radius of the cavity at the level of the liquid meniscus as r.
Neglect the change in the internal radius of the capillary tube when calculating the volume of the liquid ($\theta_c > \theta_t$)



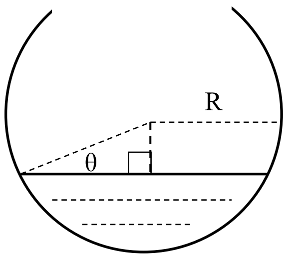
b) i) Write down the relationship between the surface energy and the surface tension coefficient of a soap film. Introduce the terms used.

ii) A child created a soap film on a circular frame of radius 3.5cm in order to make a soap bubble. What is the work that should be done by the child to convert the film of soap in to a soap bubble of radius 7cm by blowing air. Neglect the energy spent for compression of air.
Coefficient of surface tension of soap = 2.5 x 10^{-2} Nm^{-1}

iii) A liquid of coefficient of surface tension T exists in a square shaped frame in the form of a film. When a circular ring of radius r made out of an elastic material is placed on the liquid film and the section of the film inside the ring is broken, the radius of the ring increased up to R. If the coefficient of elasticity of the ring is K, show that,

$$R = r \frac{(K\pi + T)}{(K\pi - T)}$$

c) When any liquid is poured in to a small hollow spherical container of radius R, the liquid meniscus can be observed horizontally at the corresponding height of the liquid. The reason for this is the difference in contact angle made by different liquids with the wall of the container.



I. a) Copy the diagram given on to your answer sheet and mark the contact angle. Obtain an expression for the contact angle in terms of θ. (Contact angle = φ)

b) If liquids of contact angle,

1) $\varphi < 90°$ 2) $\varphi = 90°$ 3) $\varphi > 90°$

, are poured in to the above container, indicate by drawing a diagram, the position of the horizontal meniscus in each instance.

II. It was observed that there exists a small hole at the middle of the bottom surface and that when obtaining the liquid menisci of two different liquids which have the same contact angle, only one of the instances resulted in a leakage of liquid through the hole. Give 2 reasons for this.

III . A student intends to obtain the horizontal meniscus for a liquid which has a contact angle of 130° with the container. What is the minimum coefficient of surface tension the liquid should have in order to obtain a horizontal meniscus without any leakage of liquid when it is poured in to the above container which has a hole of radius 0.1mm?

Radius of the container = 10 cm Density of the liquid = 13600 kgm^{-3}

08) a)

i) Write down Newton's law of gravitation and the equation related to it. Identify each term.

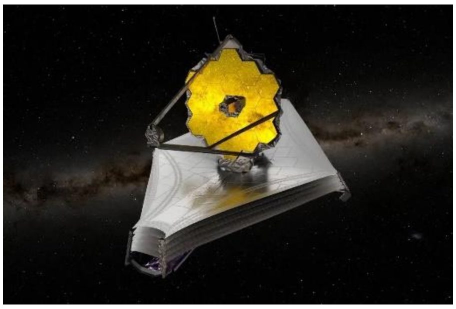
ii) a) Assuming that the Earth is a sphere of radius r and mean density d, derive an expression for the field intensity (on the surface of the Earth) in terms of d.

b) Plot the variation of the center if the Earth to infinity on a graph.

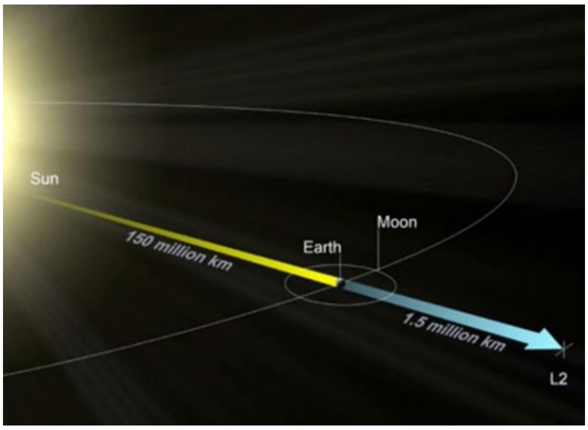
b) Answer the questions by reading the paragraph given below.

Since ancient times, man has been observing the night sky as a habit. Humans who led a nomadic lifestyle before establishing permanent settlements, observed the night sky daily with much attentiveness in order to study and comprehend the nature of stars visible in the night sky, the sun and the moon. Through this, they understood the periodic behavioral patterns of those celestial bodies. This is a special capability possessed by man which has not been gifted by nature to the other creatures. Hence, by taking the time taken for the sun to rise, set and rise again, as one 'day', ancient man obtained a certain chronological understanding of many facts such as, the moon is completely visible once in every 28 of such 'days', the motion of constellations with respect to such 'days', and regarding comets and meteorites. They were able to design calendars in this manner and were also able to predict various climatic and environmental processes which affect their survival and prepare for such disasters accordingly.

With the development of Science and Technology that occurred as a result of the findings made by various physicists and astronomers from the latter part of the Greek and Roman civilizations until today, the Telescope was discovered in order to obtain precise data regarding the outer space accurately, by going beyond the basic knowledge of the above-mentioned celestial bodies. The expansion of satellites has occurred from the development of the first telescope by Galileo Galilei in 1609 to the satellites in outer space that go beyond the Earth's surface. The Hubble Telescope, Chandra Telescope and the Kepler Space Telescope hold a special place among those.



The newest addition to the above telescopes is the James Webb Space Telescope. The life span of this telescope, which was launched on the 25th of December, 2021, is expected to be 20 years. The significance of this telescope is its location. That is, the placement of this on the point L_2, corresponding to the Sun and the Earth. This is a special position to place satellites, which was found very recently and there are mainly 2 of such positions named as Lagrange points. They are termed as L_1 and L_2, while the point closest to the Sun is L_1 and the point further away from the Sun is L_2. The significance of these points is the influence of the mass of the Earth and the mass of the Sun on the existence of these orbits.



The James Webb Telescope has a mass of about 6000 kg and it operates in the orbit of the above-mentioned L_2 point. The distance from the Earth to the Sun is around 150×10^6 km while the distance to the point L_2 from the Earth is around 1.5×10^6 km. The mass of the Earth and the Sun are about 6×10^{24} kg and 2×10^{30} kg respectively. ($G = 6.67 \times 10^{-11}\ Nm^{-2}kg^{-2}$)

i) Name 2 Telescopes which currently exist in the form of space satellites?

ii) What is the significance of Lagrange points?

iii) Show the significances of these Lagrange points (L_1 and L_2) according to your understanding. (Show separately).

iv) Develop an expression for the gravitational force acting on the James Webb Telescope and calculate the force.

v) If the point L_2 travels in a circular orbit around the Sun, calculate the angular velocity of a satellite at point L_2.

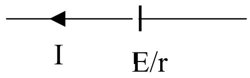
vi) Write down the gravitational potential that corresponds to point L_2.

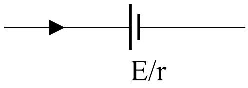
vii) Develop an expression for the work that should be done in moving the telescope from the Earth's surface to the point L_2.

Radius of the Earth	-	R
The mass of the Earth	-	M_E
The distance from the Sun to L_2	-	R_S
The distance from the Earth to L_2	-	R_E
The mass of the Telescope	-	m
The mass of the Sun	-	M_S
The distance from the Sun to the Earth	-	R_{ES}

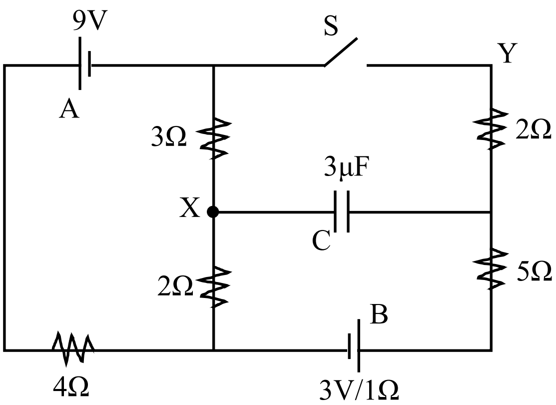
09) (A)

a) Consider a cell of electromotive force 'E' and internal resistance 'r'.

i) 

ii) 

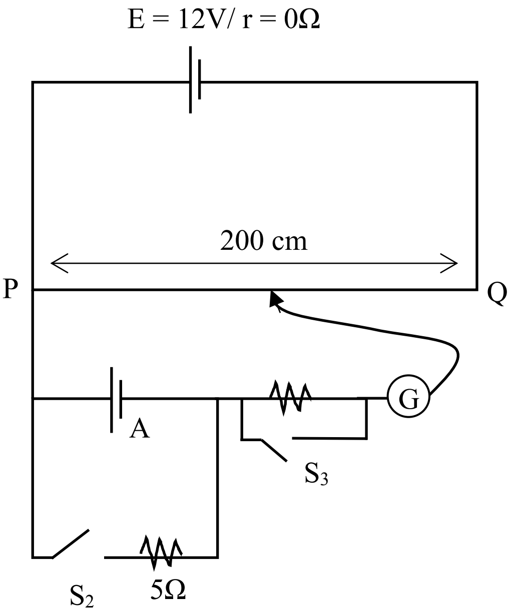
Derive expressions for the terminal potential difference of the cells in the above instances.

b) 

Consider the above circuit.

i) Calculate the current flowing through each cell when switch 'S' is closed.
ii) Calculate the charge stored in capacitor 'C' when the switch 'S' is opened and closed respectively.

c) A student says that the cell A is not ideal as expected. The circuit shown below is used to check the validity of the statement.



During the experiment 140cm and 138cm were obtained as readings.

Calculate the actual EMF and the internal resistance of the cell

09)

(B)

(a) i) Draw the circuit diagrams of each of the transistors given below

a. pnp junction bipolar transistor
b. N channel Junction Field Effect Transistor (JFET)

ii) What is the main difference with respect to the charge carries between a bipolar junction transistor and a unipolar junction transistor?

(b) The equation given below is used to find the saturated drain current (I_D) of a JFET for a given V_{GS} value.

$$I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_P}\right]^2$$

Here,

I_{DSS} = Saturated drain current

V_P = cutoff voltage

V_{GS} = Gate Source Voltage

In a certain JFET, I_{DSS} = 10mA and V_P = -4V .

1) Calculate
   i) I_D when V_{GS}= -1V
   ii) I_D when V_{GS}= -2V

2) Draw a rough sketch of the V_{GS} vs I_D graph (transconductance curve) of this transistor.

3) Draw on the same coordinate plane, the graphs of drain current (I) vs the drain source voltage (V_{DS}) when the values of V_{GS} are,

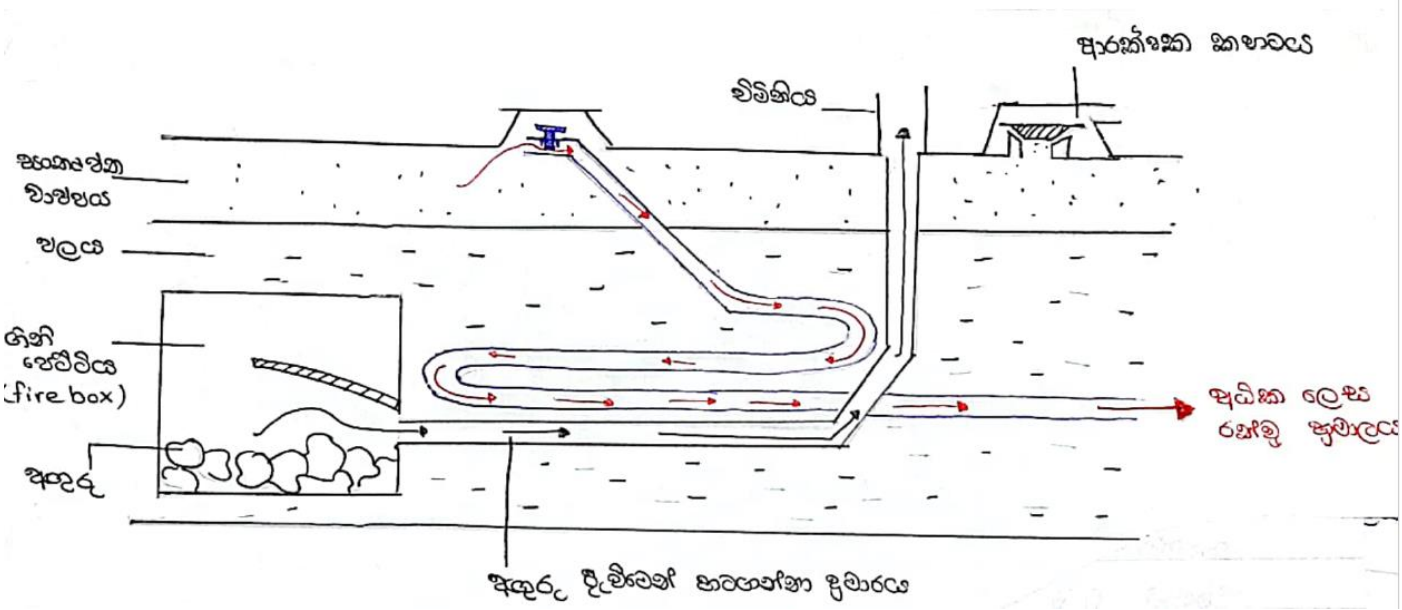
   i) 0V , ii) -1V , iii) -2V

c) For any JFET,value of the cut off voltage ,V_P is equal in magnitude to the minimum V_{DS} value which corresponds to its I_{DSS}
   i) Calculate the ohmic resistance of the above
   ii) Calculate the minimum V_{DS} value that can saturate the JFET at the instance of V_{GS}=-2V

d) Now a student decides to operate the above JFET at V_{GS}=-3V.It was found that the current through the transistor at this instance is 0.5mA

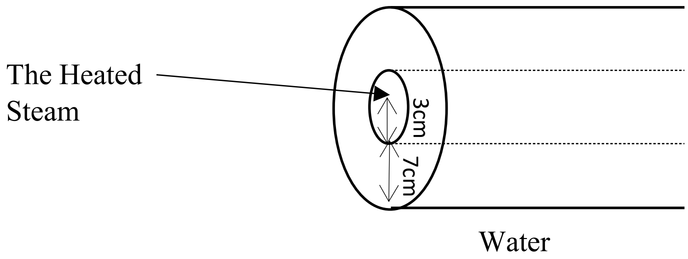
i) Using the equation introduced in (b) above or any other method identify the active mode of the transistor

10) (A)



Given above is a rough diagram of a steam engine. The super heated steam produced here is used for piston rotation. The following sums are based on this diagram.

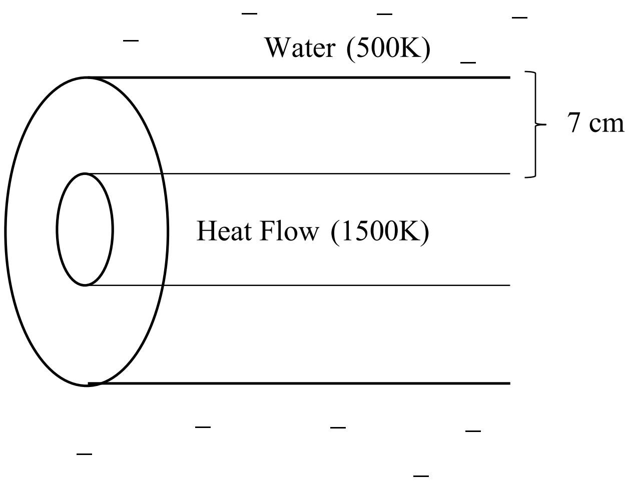
i) The heated steam produced from burning coal is sent through a network of pipes embedded in a water tank. Finally this steam is released from the chimney. Water is heated due to the fire box and the heat produced from the network of pipes.



Draw a diagram to show the direction of heat lost from the pipe.

i) Calculate the surface area through which heat enters the wall of the pipe.(length of the pipe is 1m)
ii) Calculate the surface area through which the heat leaves the pipe.
iii) Explain giving reasons, why the above heat flow is not an axial heat flow.

b)



Assume that the heat flow through the above pipe is an axial flow. Assume that the surface area through which the heat exchange occurs is equal to the mean value of surface areas calculated in (ii) and (iii) above. (The pipe does not expand due to heat flow)

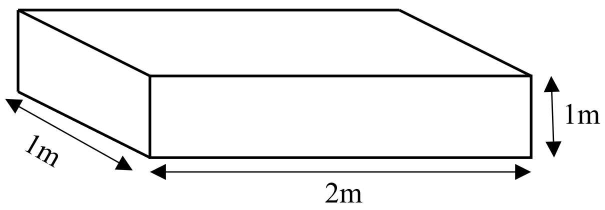
i) The thermal conductivity of the material of the above pipe (k)= 700 $Js^{-1}K^{-1}m^{-1}$. Find the rate of heat loss across the pipe mentioned above.(At the beginning of the pipe)

ii) The velocity of the steam flow through the pipe (v)=10ms^{-1} .The specific heat capacity of steam (c)=500 $Jkg^{-1}K^{-1}$. Calculate the temperature of the released steam.(Consider that the rate of rate of heat loss of the steam along the entire length of the pipe is constant and equal to the value calculated in (i) above.) (The density of steam is 2 kgm^{-3}

iii) State whether the correct temperature of the steam released from the chimney is lesser or greater than the value calculated in (ii) above.

iv) In the above setup, state the advantage of using a number of parallel pipes to carry the same amount of steam at the same velocity, than using a single pipe.

Below figure shows the water tank used in this setup.



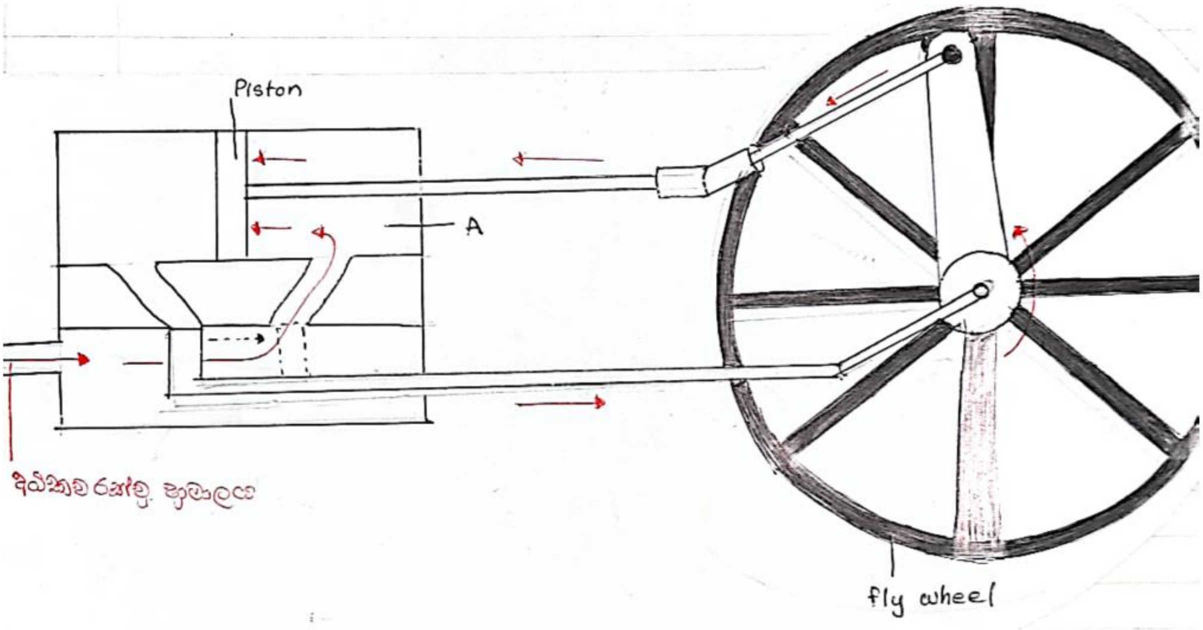
The effective volume of the tank occupied by the steam and the network of pipes including the firebox is 0.8m^3(Assume that the pipes do not expand and this volume is constant throughout the entire process) (The network of pipes and the firebox are completely submerged in water and have no contact with the water tank.)

When a beaker of water is heated by an open flame, a drop in water level is observed first. But here it is observed that the water level increases directly. Explain the reasons for this.

The initial temperature of the water tank= 27^0C (The measurements at this temperature are given in the above figure) Assume that the maximum temperature that the water can be heated up to is 500 K. Find the volume of the tank when the tank is in thermal equilibrium with water. (the coefficient of expansion of the tank is ^α& = $5x10^{-5}$ K^{-1}.

The real volume expansivity of water is $6.5x10^{-4}$ K^{-1}. Find the maximum volume of water that can be filled into the tank at 27ºCso that the water doesn't overflow and a volume of o.1 m^3 will be remaining for steam after the water is heated (500K)

d) The figure given below shows how the super heated steam produced by the engine is used to rotate wheels. The chamber A is airproof and the pistons are smooth and light. Assume that no heat exchange occurs with the environment.



i) Mention the energy conversion that occurs in the above setup.

ii) The pressure inside the chamber (P_A) is constant at $8.314x10^6$ Pa. The temperature of steam entering in to the chamber is 600K. The temperature of steam after the piston is pushed is 400K. Find out the distance that the piston moved.

(Assumptions – 1.the total energy released from the steam is used for the piston movement.
2. Steam can be considered as an ideal gas)

(Data – R= 8.314 $Jmol^{-1}K^{-1}$, The cross sectional area of the piston (A)=$2x10^{-7}$ m^3
No of moles collected in chamber A at a time= 6 mol)

10) B)

Read the passage and answer the following questions.

The primary energy source which provides energy to maintain every form of life on earth is the Sun., which is a star in the Milky Way galaxy. It consists of 74% hydrogen, 24% helium and 2% heavier elements. All these elements exist in a state called plasma, which is basically a gas in a fully ionized form. The solar atmosphere is made up of three layers. The lowest layer is the chromosphere and it has a blackbody spectrum. Most of the visible light from the sun Is radiated from the chromosphere. Corona, which is the outermost layer of the solar atmosphere continues for millions of kilometres from the photosphere. Even though it is expected to have a lower temperature in this region, it has the highest temperature in the solar atmosphere which is about 1.5 million K. This phenomenon seems to be violating the laws of thermodynamics. Scientists believe that this unexpected rise in temperature is due to releasing of energy from solar interior by its complex magnetic fields. The layers beneath the photosphere are considered as the solar interior. It has a radius of about $R_0 = 7 \times 10^8$m and a mass of about $M_0 = 2\times 10^{30}$ kg. The central
region which has an extremely high pressure and temperature is known as the core. Under these conditions the following nuclear fusion reaction occurs.

$$4H \longrightarrow He + 2e^{+} + \dot{U} + \textbf{Energy}$$

The energy created by the above reaction travels until the surface of the sun and emitted from the surface.

Wien's constant	=	3×10^{-3} mK
average distance from the sun to the earth	=	1.5×10^{11} m
mean radius of the earth	=	6.4×10^{3} m
speed of light in vacuum	=	3×10^{8} ms^{-1}
Stefan constant σ	=	6×10^{-8} Wm^{-2}K^{-4}

a) What are the three layers of the solar atmosphere?

b) What region in the solar atmosphere has the highest temperature? What can be the reason to have higher temperature in that region?

c) i) If the wavelength corresponding to the maximum intensity of radiation emitted by the photosphere is 500nm, calculate the temperature of the solar surface

ii) Luminosity (L_o) of the sun is the total energy emitted from the solar surface within a second. Calculate L_o

iii) Calculate the energy generated when four hydrogen nuclei fuses to form a helium nuclei. Hence estimate the number of reactions occurring within a second. (Mass of a hydrogen nucleus = 1.6×10^{-27}kg, mass of a helium nucleus = 6.67×10^{-27} kg, mass of positrons and neutrinos can be considered negligible.)

d)

i) The rate at which the sun's energy is received per unit perpendicular area at the top of the earth's atmosphere is defined as the solar constant (S). Calculate the solar constant.

ii) Assuming that the solar radiation falls entirely on the hemisphere facing the sun, calculate the rate of absorbance of solar energy by the earth surface

iii) If the earth is in thermal equilibrium, calculate the average temperature on the earth surface.(for objects in thermal equilibrium, the rate of energy absorption = rate of energy emission)

iv) Suggest a reason for why the above calculated value is different from the actual temperature of the earth surface.